# Decompose a Goal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

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TEACHER GUIDE · 1 OF 9

## Decompose a Goal

> Prepare Decompose a Goal: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Distinguish a **function definition** from a **function call**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/2, python-bridge/8. Reuse a familiar case for re**call** or modelling; verify that the case used to assess transfer has not already received feedback.

### Retrieval answer guidance

### Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

**Decompose** a recommender for books under 120 pages in a requested genre. A teammate scores every book before checking the page limit. Explain a clearer order and the record passed between stages.

### Concept to explain

Complex tasks become manageable when divided into testable **subproblem**s.

| Term | Meaning |
|---|---|
| decompose | Break a goal into smaller parts with clear responsibilities and connections. |
| subproblem | A smaller part of a larger problem with its own inputs and intended result. |
| dependency | Something a step needs from another step before it can proceed correctly. |
| interface | An agreed boundary specifying how two parts exchange information or actions. |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

#### Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: re**call** 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Teach **definition**, **call**, **parameter** and positional **argument** as distinct ideas. Keep **input**s scalar and avoid mutable-list aliasing in this first **function** lesson.

Reference: https://docs.python.org/3/tutorial/controlflow.html#defining-**function**s. These **function**s print results; reusable return values and local state are taught in Week 9.

**Readiness answer**

> 8. Subtraction is ordered: reversing these values would produce -8.

Python 3 editor and **function**s.py

**Parameter** cards labelled capacity and used

**Call** cards and a table for expected and observed **output**

**End goal for the pictorial case**

By the end, I can **decompose** a calculation into a **function** with explicit **argument**s and a returned result.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Decompose a Goal using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

**Input → Process → Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Goal: help a learner leave home ready.

2. Parts: check time; check weather; check bag; choose reminder.

3. **Dependency**: weather must be known before the clothing suggestion.

4. Test each part separately, then test the whole flow.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

**Concrete teaching case**

### The teaching case

A class wants to recommend one of five library books.

### The teaching case

**Decompose** the goal into checkable steps.

### Exact explanation to model

> **Why the result follows**
>
> Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

### Explain the added worked example

## Explain the added worked example · 1

A fictional supply cupboard records capacity and the number of occupied spaces. The remaining-space calculation is capacity minus used.

Instead of copying the same instructions for every cupboard, define one **function** and **call** it with different values. A **function** makes a procedure reusable; its name alone does not guarantee that its instructions are correct.

Our **input** contract requires nonnegative integer counts with used no greater than capacity. The example assumes these conditions; it does not validate them. This week the **function** prints its result.

Week 9 will show how to return a value for later calculations.

## Explain the added worked example · 2

def introduces the **definition**. The **function** name is followed by **parameter** names in parentheses and a colon. Four-space indentation marks the body. Executing the **definition** creates the **function** but does not run its body.

The two unindented **call**s then run it, producing 8 and 3 on separate lines. A blank line helps readability; indentation determines the block.

### Key terms

| Term | Meaning |
| --- | --- |
| Function | A named block of reusable instructions that runs when called |
| Definition | Code that creates a function and specifies its body |

| Term | Meaning |
| --- | --- |
| Call | An instruction that runs a function with supplied arguments |
| Parameter | A name in a function definition that receives an argument |
| Argument | A value supplied when calling a function |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Execution event | What happens |
| --- | --- |
| Execute the definition | Create the function; no output yet |
| Call with 12 and 4 | Match parameters; subtract; print 8 |
| Call with 10 and 7 | Match new values; subtract; print 3 |
| After the second call | Continue with the next statement |

## Model the reasoning aloud · 3

In show_remaining(12, 4), 12 and 4 are **argument**s; capacity and used are **parameter**s.

Positional **argument**s match from left to right. show_remaining(4, 12) produces -8, showing why order matters; it also violates our written **input** contract. Use a fresh file with the **definition** before its **call**s.

> **Predict the output of show_remaining(9, 9) followed by show_remaining(9, 0). Name the parameter receiving each argument. What would be printed if both calls were removed?**

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Decompose a Goal

A fictional library computes total spine labels needed for several equal shelves.

### The facts we will use

#### Supplied facts

Define labels(shelves,per_shelf): return shelves*per_shelf.

#### Supplied facts

**Call** needed=labels(4,6), then print(needed). No other side effects are defined.

AI ACADEMY / AI-2/2

# Decompose a Goal

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Call with arguments

The call supplies shelves=4 and per_shelf=6 locally.



### 2. Function body

Multiply 4 x 6 to calculate 24.



### 3. Return value

return sends 24 back to the caller; it does not print.



### 4. Caller prints

needed receives 24; print(needed) displays 24.

**CHANGE ONE THING**

Call labels(4,0) instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Call with arguments | The call supplies shelves=4 and per_shelf=6 locally. |
| 2. Function body | Multiply 4×6 to calculate 24. |
| 3. Return value | return sends 24 back to the caller; it does not print. |
| 4. Caller prints | needed receives 24; print(needed) displays 24. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows


### Why the result follows


A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

### What this example does not establish

#### Keep the stated limit

This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

### Change one thing in the pictured case

#### Change one thing

Change: **Call** labels(4,0) instead.

#### Predicted result

**Prediction**: The returned result is 0, and the **call**er prints 0.

#### Why the result follows

Reason: Four shelves times zero labels per shelf is zero.

### Ask while showing the picture

## Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Decompose a Goal, reduce support deliberately and check individual reasoning before discussing answers.

## Board sequence

### 1. Put the facts on the board

Write the given facts: A class wants to recommend one of five library books.

### 2. State the decision to explain

Write the question to explain: **Decompose** the goal into checkable steps.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: Collect preferences, record book traits, score each with a declared rule, handle ties and explain the suggestion. Each step produces an inspectable record.

### Hold back the independent task

Keep for an independent attempt: A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

## Classroom dialogue

### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

The **call**s print 0 then 9. capacity receives 9 in both **call**s; used receives 9 then 0. Removing both **call**s leaves a **definition** with no printed **output**. Exit: show_remaining(10, 7) should print 3 but the faulty version prints 8.

Passing two cases does not establish correctness for all allowed **input**s or validate the **input** contract.

### Case-specific teaching sequence

| Stage | Teacher action |
|---|---|
| Session 1 start to 7 minutes | Retrieve ordered subtraction with the prerequisite. |
| 7 to 20 minutes | Read the contract and label the definition and calls for W1. |
| 20 to 33 minutes | Match argument cards to parameters and trace the worked calls. |
| 33 to 45 minutes | Complete W2 and the guided question before running the file. |
| Session 2 start to 8 minutes | Retrieve why a definition alone prints nothing. |
| 8 to 20 minutes | Run the fixed-number bug, complete W3 and rerun the earlier case after repair. |
| 20 to 35 minutes | Collect the independent W4 function and predictions before hints. |
| 35 to 45 minutes | Discuss W5 and use the exit to assess the limits of passing tests. |

### Reduce help deliberately

#### Completion practice

Completion task: Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

#### Reasoning to listen for

Teacher check: The **call**s print 5, 0 and 8 on separate lines. A **definition** alone creates the **function** but prints nothing because its body has not been **call**ed.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

#### Expected repair

Expected correction: The **definition** creates the **function**; a **call** runs its body.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

> What happens visibly if the **call**er stores needed but omits print(needed)?

### Reasoned teacher answer

The calculation returns 24 and needed stores it, but the shown program displays nothing because no print runs.

Changed-case reasoning: The returned result is 0, and the **call**er prints 0. Four shelves times zero labels per shelf is zero.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the weekly evidence lab

> Run the Decompose a Goal investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A **definition** matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Cut a familiar task into 4–6 cards. Order dependencies, name each **input** and **output**, and test whether replacing one card breaks the whole system.

> What did the hardest test teach you that the normal test did not?

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

Writing def runs the body immediately: The **definition** creates the **function**; a **call** runs its body.

**Argument**s and **parameter**s are interchangeable labels: **Parameter**s are names in the **definition**; **argument**s supply values at a **call**.

Swapping **argument**s never matters: Different **parameter** roles can give different results when values are swapped.

A correct first example proves the **function** is reusable: Fixed numbers can match one case while ignoring later **argument**s.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Decompose a Goal questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

**Teacher key · Worked case answer**

Collect preferences, record book traits, score each with a declared rule, handle ties and

explain the suggestion. Each step produces an inspectable record.

### Independent question

#### Independent question · withhold the key

A recommender must receive a topic, find matching books and show one title. Split the goal into three tasks and state what passes between them.

### Expected independent answer

#### Teacher key · Expected independent answer

Collect a topic; use that topic to select matching records; pass a chosen record to display its title. Each step has a specific **input** and **output**.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “Decomposition is only making a longer task list.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

Read request; validate required fields; filter by genre and pages<120; score remaining books; apply a stated tie/no-match policy; explain the result. Pass request, filtered IDs, scores and selected ID between stages.

Scoring first is not necessarily wrong if filtering still occurs, but it makes the constraint easier to overlook.

### Explanations for the additional workbook examples

### W1 Label the parts

#### Workbook question · W1

In def show_remaining(capacity, used), identify the **function** name and **parameter**s. In show_remaining(15, 6), identify the **argument**s and their matching **parameter**s.

#### Teacher reasoning · W1

Name show_remaining; **parameter**s capacity and used. **Argument**s 15 and 6 match capacity = 15 and used = 6. Expected printed result 9.

### W2 Follow the execution order

#### Workbook question · W2

Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

#### Teacher reasoning · W2

The **call**s print 5, 0 and 8 on separate lines. A **definition** alone creates the **function** but prints nothing because its body has not been **call**ed.

### W3 Repair ignored arguments

#### Workbook question · W3

The faulty body calculates 12 - 4 regardless of the **call**. State the **output** and expected **output** for show_remaining(11, 2). Repair the expression and name an earlier **call** to rerun.

#### Teacher reasoning · W3

Faulty **output** 8; expected 9. Use capacity - used. Rerun show_remaining(12, 4), expecting 8, as well as the revealing **call**.

### W4 Create a new function independently

#### Workbook question · W4

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

### Teacher reasoning · W4

The body prints boxes * 6 + loose. **Output**s are 20, 2 and 18. show_stock(2, 3) gives 15 because boxes and loose have different roles. Define once and use three **call**s; do not hard-code one answer.

**W5 State the input assumption**

### Workbook question · W5

The correct subtraction **function** prints -1 for show_remaining(4, 5). Does successful execution make this a valid cupboard record? Explain what the **function** assumes and what needs checking.

### Teacher reasoning · W5

No. used exceeds capacity and violates the contract. The **function** calculates with supplied values but does not check the contract. A separate validation check or guarded **function** is needed to reject invalid records.

**Assess independent transfer**

### Independent question · withhold the key

Selected case: Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

### Selected case answer

> **Teacher key · Selected case answer**
>
> The body prints boxes * 6 + loose. **Output**s are 20, 2 and 18. show_stock(2, 3) gives 15 because boxes and loose have different roles. Define once and use three **call**s; do not hard-code one answer.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automati**call**y an equivalent test.

## Connect and extend learning

> Connect Decompose a Goal to Transparent Book Recommender, adapt support and plan a later changed-case check.

### Transparent Book Recommender

Separate reader profile, book data, score calculation, ranking, and explanation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 3: Branching Logic. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Use two labelled **parameter** spaces and place **argument** cards left to right. Permit an oral execution trace alongside code.

Extension: add an if check inside show_remaining that prints Invalid when used exceeds capacity, otherwise prints the difference; test (4, 5) and (4, 4). This checks only that one constraint.

W4 must define one **function**, use both **parameter**s, **call** it three times and predict 20, 2, 18; the swapped case must give 15.

After support, retry show_tiles(packs, singles) with 4 tiles per pack: **call**s (2, 3), (0, 3), (2, 0) print 11, 3, 8; swapped (3, 2) prints 14. Record assistance and collect new **prediction**s before execution.

Week 9 returns results to the **call**er and explains local variables and fresh state on each **call**.

### Plan the delayed return

Suggested return points in this level: ai-2/3, ai-2/5, ai-2/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
|---|---|
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently.

## Prepare, teach and assess

> Use this preparation summary alongside the detailed Decompose a Goal lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Distinguish a **function definition** from a **function call**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order.

Explain what happens if the file contains only the **definition**.

### Supported practice answer

Reasoned practice answer: The **call**s print 5, 0 and 8 on separate lines. A **definition** alone creates the **function** but prints nothing because its body has not been **call**ed.

Check every learner before discussion: Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: The **definition** creates the **function**; a **call** runs its body.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: **Call** labels(4,0) instead. Require a **prediction** and a reason before comparing. Expected result: The returned result is 0, and the **call**er prints 0. Reason: Four shelves times zero labels per shelf is zero.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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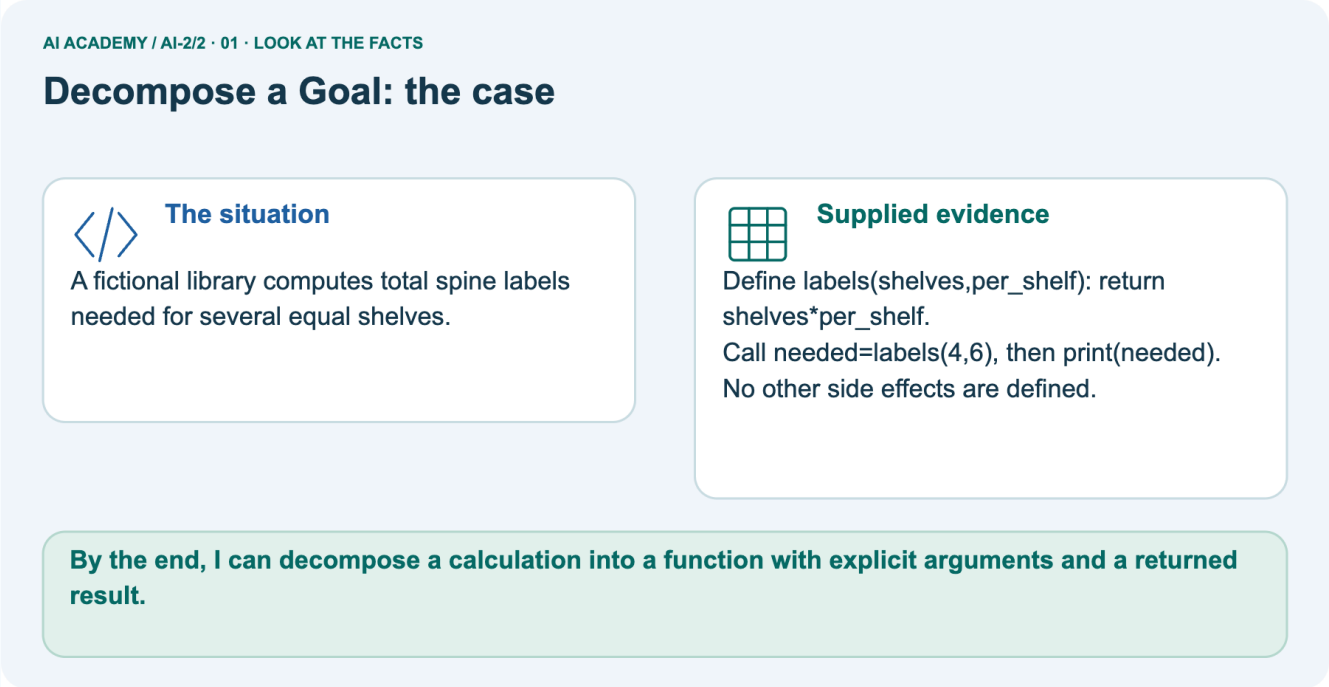


## Visual concept journey

> Show one Decompose a Goal graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Decompose a Goal: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional library computes total spine labels needed for several equal shelves. Supplied facts: Define labels(shelves,per_shelf): return shelves*per_shelf. **Call** needed=labels(4,6), then print(needed).

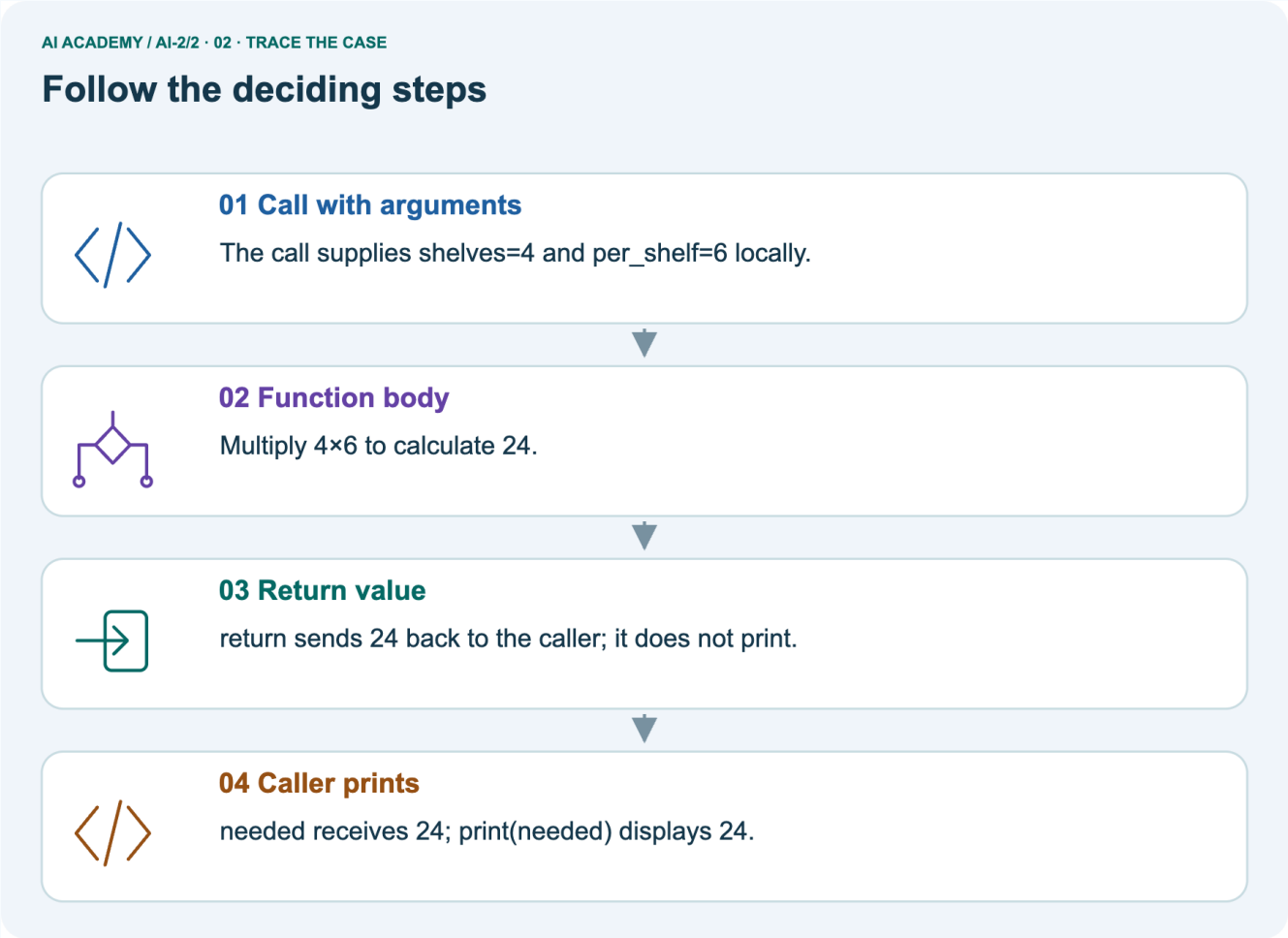
No other side effects are defined.

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

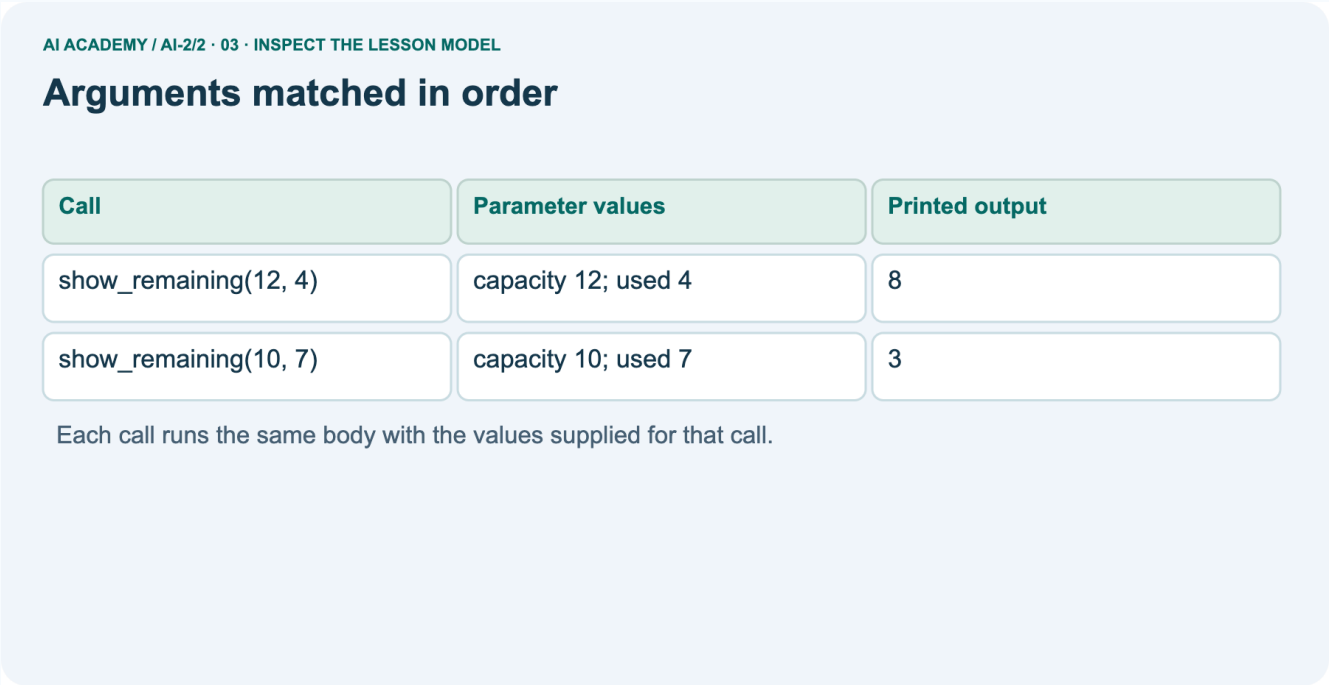
**Call** with **argument**s: The **call** supplies shelves=4 and per_shelf=6 lo**call**y. → **Function** body: Multiply 4×6 to calculate 24. → Return value: return sends 24 back to the **call**er; it does not print. → **Call**er prints: needed receives 24; print(needed) displays 24.

> Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Arguments matched in order




Follow the labelled picture with the matching explanation. Open the vector diagram

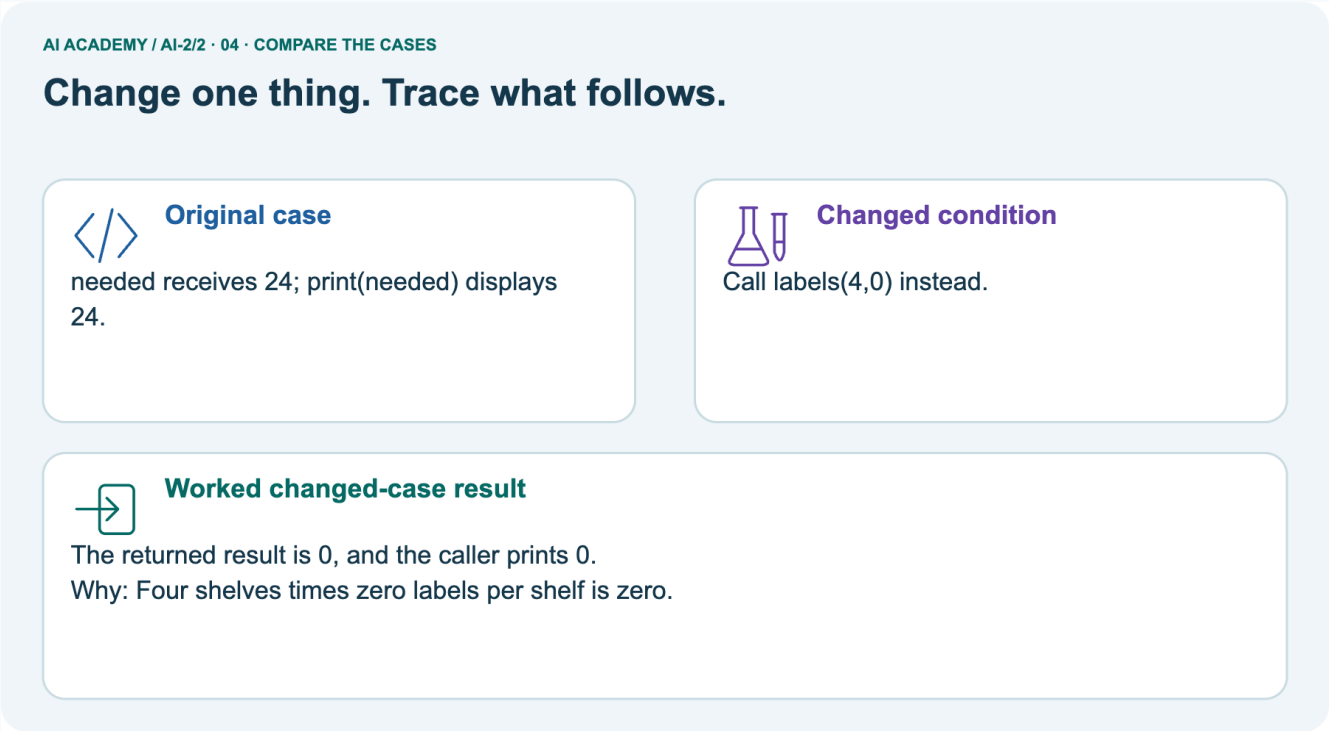
**Argument**s matched in order. show_remaining(12, 4) | capacity 12; used 4 | 8. show_remaining(10, 7) | capacity 10; used 7 | 3

Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: needed receives 24; print(needed) displays 24. Changed condition: **Call** labels(4,0) instead. Expected result: The returned result is 0, and the **call**er prints 0.

Reason: Four shelves times zero labels per shelf is zero.

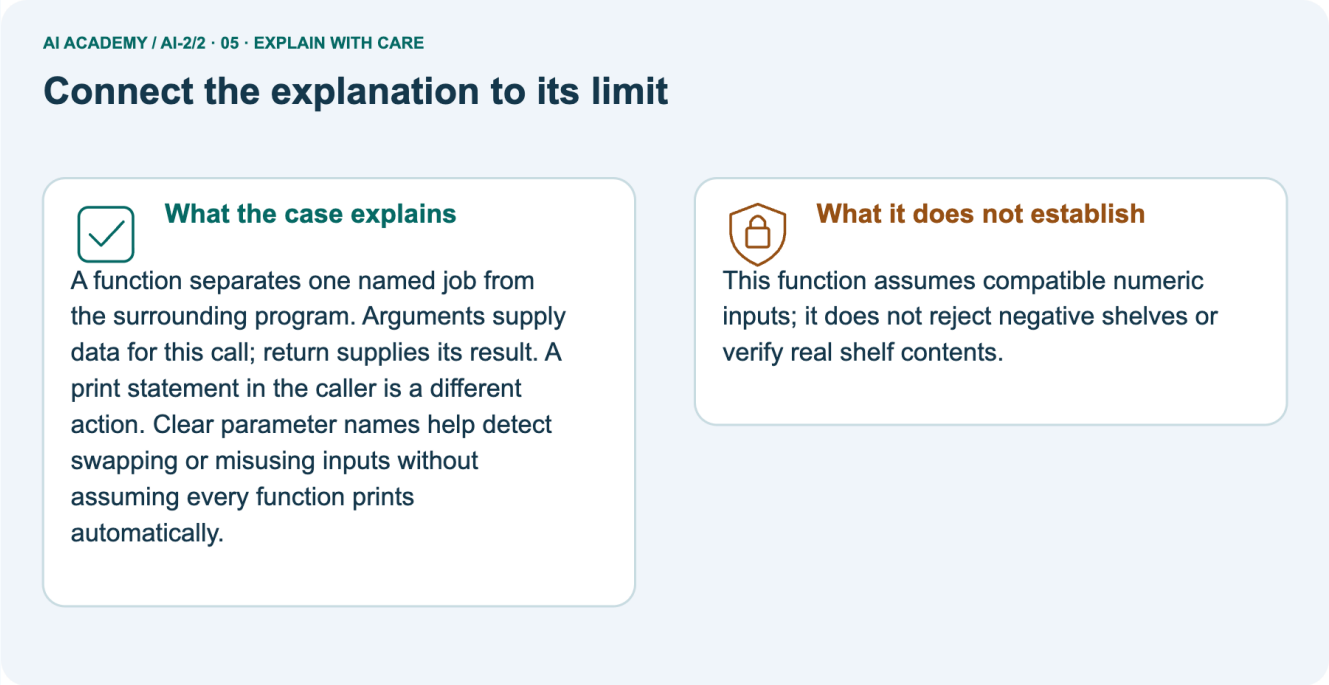
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Four shelves times zero labels per shelf is zero.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

A **function** separates one named job from the surrounding program. **Argument**s supply data for this **call**; return supplies its result. A print statement in the **call**er is a different action.

Clear **parameter** names help detect swapping or misusing **input**s without assuming every **function** prints automati**call**y.

Limit: This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

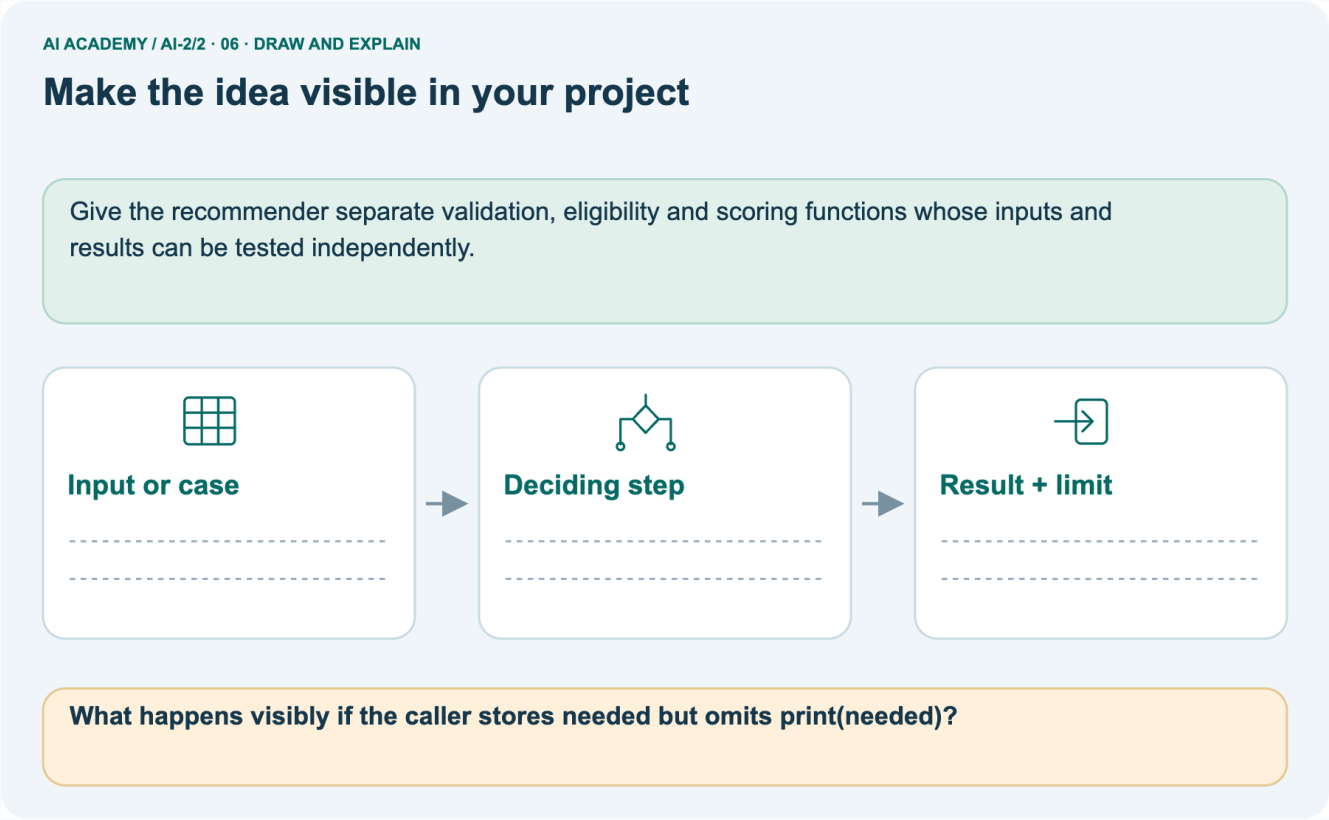
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This **function** assumes compatible numeric **input**s; it does not reject negative shelves or verify real shelf contents.

### 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender separate validation, eligibility and scoring **function**s whose **input**s and results can be tested independently. Draw your **input**, deciding step, result and limit.

Supported question: What happens visibly if the **call**er stores needed but omits print(needed)?

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The calculation returns 24 and needed stores it, but the shown program displays nothing because no print runs.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.